

EDUCATION

• Purdue University, West Lafayette	2023 - Present
Ph.D. in Mechanical Engineering	GPA: 3.66/4.0
• University of Michigan, Ann Arbor	2020 - 2022
M.S. in Robotics	GPA: 3.92/4.0
• Purdue University, West Lafayette	2014 - 2020
B.S.E. in Mechanical Engineering	GPA: 3.76/4.0

PROFESSIONAL EXPERIENCE

• Graduate Teaching Assistant , Purdue University, West Lafayette	2025 - Present
• Graduate Research Assistant , Purdue University, West Lafayette	2023 - Present
• Research Associate , University of Michigan, Ann Arbor	2022 - 2023
• Graduate Research Assistant , University of Michigan, Ann Arbor	2021 - 2022
• Undergraduate Research Assistant , Purdue University, West Lafayette	2019 - 2020

JOURNAL PUBLICATIONS

‡: Equal contribution

*: Corresponding author

- Lee, H., Sim, Y., Han, C., Jun, M., & Lee, J.* (2025). GRAVITON++: Graph-based Robust Resampling Algorithm for Visual Inference in Target Object Localization and Certification [In preparation].
- Lee, H., Sim, Y., Han, C., Jun, M., & Lee, J.* (2025). BRIDGES: Bi-directional Robotic Integration with Digital-twin for Generalized, Efficient, and Safe Automation in Manufacturing [In preparation].
- Sim, Y., Lee, J., Lee, H., Yun, D., Myeong, N., Hwang, Y., Park, H., Jun, M., & Kim, E.* (2025). Sound Data Augmentation Using Frequency Response Superposition for Machine Tool State Recognition. Journal of Manufacturing Science and Engineering [In preparation].
- Lee, H., Han, C., Gabor, T., Sim, Y., Akin, S., Jun, M., & Lee, J.* (2025). Manufacturable Physically Unclonable Identifiers via Cold Spray: A Full-Stack Framework with Implicit Neural Representation for Scalable Product Authentication. Advanced Engineering Informatics [Under review].
- Kim, E., Lee, H., Sim, Y., Lee, J., & Jun, M.* (2025). Overcoming Sparse Run-to-Failure Data Challenges in Manufacturing: A Contrastive Mixer Framework for Remaining Useful Life Prediction. CIRP Annals – Manufacturing Technology.
- Lee, H., Sim, Y., Han, C., Jun, M., & Lee, J.* (2025). GRAVITON: Generalized Resampling Algorithm for Visual Inference in Target Object Localization. Computers in Industry [Under revision].
- Han, C., Gabor, T., Lee, H., Lee, J., Akin, S., & Jun, M.* (2025). Pulsed Cold Spray System for Physical Unclonable Function Generation. Procedia CIRP.
- Jeon, J., Sim, Y., Lee, H., Han, C., Yun, D., Kim, E., Nagendra, S. L., Jun, M., Kim, Y., Lee, S., & Lee, J.* (2025). ChatCNC: Conversational Machine Monitoring via Large Language Model and Real-Time Data Retrieval Augmented Generation. Journal of Manufacturing Systems, 79, 504-514.
- Lee, H.*, Toner, T., Tilbury, D., & Barton, K. (2025). GraspMixer: Hybrid of Contact Surface Sampling and Grasp Feature Mixing for Grasp Synthesis. IEEE Transactions on Automation Science and Engineering.
- Lee, J., Akin, S.*, Sim, Y., Lee, H., Kim, E., Nam, J., Song, K., & Jun, M.* (2024). A stethoscope-guided interpretable deep learning framework for powder flow diagnosis in cold spray additive manufacturing.

Manufacturing Letters.

- Han, C., Lee, J., **Lee, H.**, Sim, Y., Jeon, J., & Jun, M.* (2024). Zero-shot Autonomous Robot Manipulation via Natural Language. Manufacturing Letters, 42, 16-20.
- **Lee, H.***, Toner, T., Tilbury, D., & Barton, K. (2022). Multi-sensor aided deep pose tracking. IFAC-PapersOnLine, 55(37), 326-332.

CONFERENCE PROCEEDINGS & PRESENTATIONS

‡: Presenter

*: Corresponding author

- Han, C.‡, Sim, Y., Jung, H., Lee, J., **Lee, H.**, Kang, Y., Woo, S., Kim, G., Park, H., & Jun, M.* (2025, December 2-7). IMPACT: Industrial Machine Perception via Acoustic Cognitive Transformer. Conference on Neural Information Processing Systems 2025: San Diego, CA, United States [Under review].
- **Lee, H.‡***, Sim, Y., Han, C., Lee, J., Bera, A., & Jun, M. (2025, October 19-25). EASEIR: Efficient and Adaptive Safe-set Estimation via Implicit Representation for High-dimensional Motion Planning. IEEE/RSJ Conference on Intelligent Robots and Systems 2025: Hangzhou, China [Accepted].
- Kim, E., **Lee, H.**, Sim, Y., Lee, J., & Jun, M.‡* (2025, August 17-23). Overcoming Sparse Run-to-Failure Data Challenges in Manufacturing: A Contrastive Mixer Framework for Remaining Useful Life Prediction. 74th CIRP Annals – Manufacturing Technology: Stockholm, Sweden.
- Sim, Y.‡*, Lee, J., **Lee, H.**, Han, C., Lee, H., & Jun, M. (2025, August 17-20). I2DTM: Immersive and Interactive Digital Twin Metaverse Framework. International Design Engineering Technical Conference & Computers and Information in Engineering Conference 2025: Anaheim, CA, United States [Under review].
- Sim, Y.‡*, Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., Jun, M., & Kim, E. (2025, June 23-27). Sound Data Augmentation Using Frequency Response Superposition for Machine Tool State Recognition. International Manufacturing Science and Engineering Conference 2025: Greenville, NC, United States.
- **Lee, H.‡***, Choi, Y., Lugo, E., Lee, J., Lee, S., & Jun, M. (2025, June 23-27). GMGP: Generalized Model for Grasp Planning of Vacuum and Parallel Jaw Grippers. International Manufacturing Science and Engineering Conference 2025: Greenville, NC, United States.
- Han, C.‡, Gabor, T., **Lee, H.**, Lee, J., Akin, S., & Jun, M.* (2025, June 1-4). Pulsed Cold Spray System for Physical Unclonable Function Generation. CIRP International Symposium on Electro Physical and Chemical Machining (ISEM): Vancouver, BC, Canada.
- Jun, M.‡*, Lee, J., Kim, E., Sim, Y., Han, C., & **Lee, H.** “Generalized and Generative AI for Smart Manufacturing” Presented at International Conference on Precision Engineering and Sustainable Manufacturing, July 2024
- Han, C.‡, **Lee, H.**, Lee, J., & Yun, H.* (2024, June 17-21). Digital Twins for Autonomous CNC Equipment Operation. International Manufacturing Science and Engineering Conference 2024: Knoxville, TN, United States.
- **Lee, H.‡***, Toner, T., Tilbury, D., & Barton, K. Multi-sensor aided deep pose tracking. (2022, October 2-5). Modeling, Estimation and Control Conference: Jersey City, NJ, United States.

POSTER SESSIONS

- Sim, Y., Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., Jun, M., & Kim, E. “Sound Data Augmentation Using Frequency Response Superposition for Machine Tool State Recognition.” Presented at Manufacturing And Materials Research Laboratories Symposium, December 2024
- Lee, J., Jeon, J., Sim, Y., **Lee, H.**, Han, C., Yun, D., & Jun, M. “Generative AI Based Real-Time Human-Data Interaction for Smart Manufacturing” Presented at Digital Twin for Manufacturing Sustainability, Safety, and Resilience, October 2024

- **Lee, H.**, Sim, Y., Lee, J., Han, C., & Jun, M. “Digital Twin & Vision Guided Autonomous Robotic Path Planning” Presented at Digital Twin for Manufacturing Sustainability, Safety, and Resilience, October 2024
- Lee, J., Kim, E., Han, C., **Lee, H.**, & Sim, Y. “Generalized and Generative Smart Manufacturing” Presented at North Academy of Engineering Regional Meeting, March 2024
- Han, C., **Lee, H.**, & Jun, M. “Robotics in Smart Manufacturing” Presented at Congressional Artificial Intelligence Caucus, May 2024
- **Lee, H.** & Toner, T. “Multi-sensor aided deep pose tracking.” Presented at Toyota HSR Community in North America Meeting 3, May 2022

FELLOWSHIPS

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| • Winkelman Fellowship | 2023 - 2024 |
| • Multidisciplinary Design Program (MDP) Summer Fellowship | 2021 - 2021 |

OTHER HONORS & AWARDS

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| • Italian Technology Award 2025 | N/A |
| • Young Professional NET Award 2025 | N/A |
| • NSF Travel Award for NAMRC/MSEC 2025 | N/A |
| • DigiTwin 2024 for Manufacturing, Sustainability, Safety, and Resilience Student Best Poster Award | N/A |
| • Young Professional NET Award 2024 | N/A |
| • MECC 2023 Travel Grant | N/A |

PATENTS

- Jun, M., Lee, J., **Lee, H.**, Akin, S., Han, C., Gabor, T., & Sim, Y. “Cold Spray-Enabled Physically Unclonable Identifier and its Spectral Authentication via Implicit Neural Representation.” U.S. Patent. [Under review]

INDUSTRIAL PROJECTS

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| • BCI Solution, Inc | 2024 - 2025 |
| ○ Developed a digital twin of FANUC CRX-10iA/L using MuJoCo physics simulator and Python | |
| ○ Established bi-directional communication among FANUC CRX-10iA, its digital twin, and CESMII platform for automation and verification of core-making operations | |
| • Korea Institute of Machinery and Materials (KIMM) | 2024 - Present |
| ○ Developed a generalized sampling algorithm to extend a cutting edge object segmentation AI to 6-DOF object pose estimation method for machine tending | |
| ○ Developed a hybrid computer vision pipeline to localize metal debris inside CNC machine that must be removed | |
| • BCI Solution, Inc | 2023 - Present |
| ○ Developed a model predictive control (MPC) algorithm using Python to plan a path for FANUC CRX-25iA to apply sealant on target locations | |
| ○ Built a server-client system using Python and KAREL to communicate FANUC controller using external PC | |
| • Terra Drive Systems (TDS) | 2023 - 2023 |
| ○ Installation of stethoscope-based sound sensor and mini PC to establish a data pipeline | |
| ○ Developed a sound recognition AI model to monitor status of Makino a81 CNC machine | |
| ○ Generated a web-based dashboard for intuitive data analysis and visualization | |

PROFESSION SOCIETY MEMBERSHIPS

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| • Korean-American Scientists and Engineers Association (KSEA) / Member | 2024 - Present |
| • Graduate Association for Manufacturing Excellence / Treasurer | 2025 - Present |

MINI PROJECTS

- Robot-Camera Pose Tracking 2022 - 2022
 - Preprocessed and trained Keypoint-RCNN using PyTorch to track and visualize all joint positions of Franka Emika Panda in RGB images
 - Integrated forward kinematics, Perspective-n-Point algorithm, and camera intrinsic matrix to estimate camera poses with respect to the robot's base
- Picking & Sorting (5-DOF Robotic Arm) 2021 - 2022
 - Utilized OpenCV to detect blocks using LiDAR Camera L515 (Intel RealSense)
 - Implemented forward and inverse kinematics of a robot manipulator for toy block picking, sorting, and stacking
- SLAM (Differential Wheeled Robot) 2021 - 2021
 - Embedded fractions of particle filter algorithm in C for localization and mapping using LiDAR sensor
 - Generated A* search path planning algorithm in C to automatically explore unknown environments
 - Constructed a control strategy in C to integrate several sensors to make the robot follow the planned path
- Adaptive Yaw Rate Controller Design 2020 - 2020
 - Explored methods to obtain an optimal performance for vehicle system parameter estimation
 - Designed a controller using Indirect Model Reference Adaptive Control (IMRAC)
 - Simulated the performance of the controller under noise, disturbance, and model uncertainties using Simulink
- Line Tracking Robot 2019 - 2019
 - Designed PI controller using LabVIEW and MATLAB to control 2 motors under a robot
 - Utilized infrared line tracking sensor as a part of feedback system for stabilizing the robot control
- MOTHERSHIP (Serpentine Rescue Robot) 2015 - 2016
 - Analyzed structural failures in articulators of a serpentine robot during its operation
 - Explored alternative structures, mechanisms, and materials to prevent cracking in the joints

LEADERSHIP & MENTORSHIP

- Undergraduate Student Mentor 2024 - Present
 - Richard Li (Summer 2024), Evin Lugo (Fall 2024 - Present), Yoonseo Lee (Spring 2025 - Present)
- Sub-Team Leader (Barton Research Group) 2021 - 2022
 - Facilitated communication among 5 sub-team members and Ph.D. mentors
 - Led weekly team meetings for discussion among members and provided advice regarding their projects
 - Helped the members to view their individual works in a broader aspect and tied them together
- Squad Leader (Republic of Korea Army) 2016 - 2018
 - Trained artillery specialty to new members and helped them understand their duty in a team
 - Fostered cohesion among members by discussing and reminding them about team goals

CERTIFICATIONS

- Deep Learning Specialization, DeepLearning.AI
- Modern Robotics, Northwestern University

SKILLS

- **Programming:** C/C++ (Beginner), Python (Professional), MATLAB (Professional)
- **Development:** Linux, Git, Visual Studio Code, Fusion 360
- **Frameworks/Libraries:** ROS, PyTorch, TensorFlow, OpenCV, Open3D, Arduino
- **Languages:** English (bilingual), Korean (native), Japanese (conversational)